

Parameter recovery: acuity  
selection on the tested grid

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Parameter recovery probes whether the executed observation model contains enough information to distinguish sensor acuity under the study design. We sweep acuity values 0.60, 0.70, 0.80, 0.90:

# Parameter recovery (part 2)

for each true acuity the model generates 200 synthetic observations per trial across 960 independent trials, fits acuity by marginal-likelihood grid search, and compares the recovered value with ground truth.

# Parameter recovery (part 3)

The configured seed initializes the base RNG stream; it is not the independent analysis unit.

# Parameter recovery (part 4)

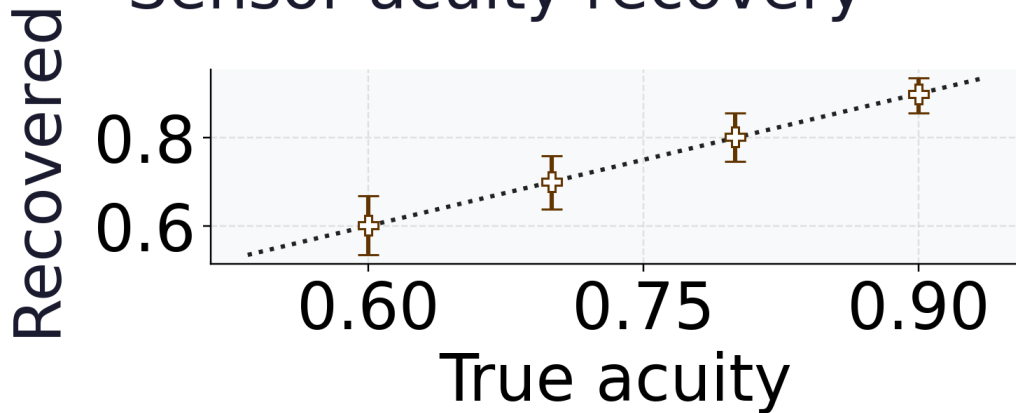
Across the sweep the mean absolute recovery error is 0.0232 and the coefficient of determination of mean-recovered versus true acuity is  $R^2 = 0.9999$ . Within this finite grid and observation budget, recovered acuity tracks the identity line with the reported error (fig. 31).

# Parameter recovery (part 5)

This is evidence of practical acuity recoverability for the executed design, not a proof of global or structural identifiability and not an acuity-by-colony-size study.

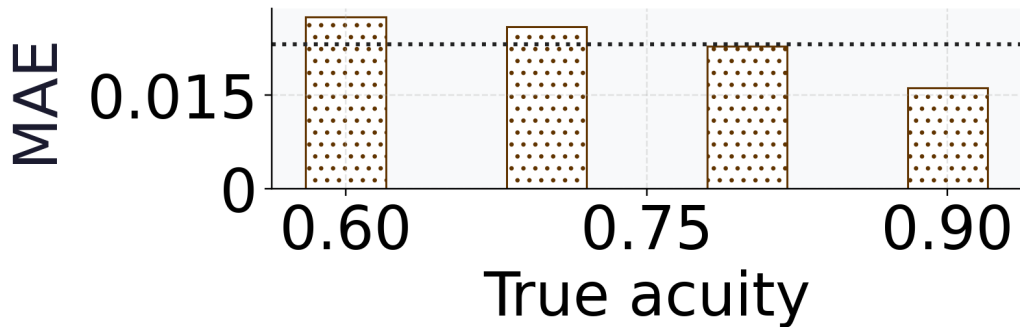
# Parameter recovery (part 6)

## Sensor acuity recovery



# Parameter recovery (part 7)

Absolute recovery error



# Parameter recovery (part 8)

Recovery whiskers: 95% empirical percentile intervals over independent simulation trials within each acuity. Observations are nested within trial.

# Parameter recovery (part 9)

independent trials/acuity = 960  
observations/trial = 200  
MAE = 0.0232

# Parameter recovery (part 10)

$$R^2 = 0.9999$$

# Parameter recovery (part 11)

Simulation-based recovery over the declared acuity grid. Identity reference is not an additional estimate; no uncertainty interval is inferred for MAE bars.